

1. Prove the following elementary results

$$\int a dx = ax + c, \quad \int x dx = \frac{x^2}{2} + c, \quad \int x^m dx = \frac{x^{m+1}}{m+1} + c \quad m \in \mathbb{N}$$

$$\int \frac{dx}{x^2} = -\frac{1}{x} + c \quad x \neq 0, \quad \int \frac{dx}{2\sqrt{x}} = \sqrt{x} + c \quad x \neq 0$$

$$\int x^\alpha dx = \frac{x^{\alpha+1}}{\alpha+1} + c \quad \alpha \in \mathbb{Q} - (-1), \quad \int \frac{dx}{x} = \ln|x| + c \quad x \neq 0$$

$$\int e^x dx = e^x + c, \text{ if } a > 0 \text{ and } a \neq 1, \quad \int a^x dx = \frac{a^x}{\ln a} + c \quad x \neq k\pi \quad k \in \mathbb{Z}$$

$$\int \sin x dx = -\cos x + c, \quad \int \cos x dx = \sin x + c$$

$$\int \frac{dx}{\cos^2 x} = \int (1 + \tan^2 x) dx = \tan x + c \quad k \in \mathbb{Z}, \quad \int \frac{dx}{\sin^2 x} = -\cotan x + c$$

$$\int \operatorname{sh} x dx = \operatorname{ch} x + c, \quad \int \operatorname{ch} x dx = \operatorname{sh} x + c$$

$$\int \operatorname{th} x dx = \ln(\operatorname{ch} x) + c, \quad \int \frac{dx}{\operatorname{ch}^2 x} = \int (1 - \operatorname{th}^2 x) dx = \operatorname{th} x + c$$

$$\int \cos^2 x dx = \frac{x}{2} + \frac{\sin 2x}{4} + c, \quad \int \sin^2 x dx = \frac{x}{2} - \frac{\sin 2x}{4} + c \quad x \neq \frac{\pi}{2} + k\pi \text{ and } k \in \mathbb{Z}$$

$$\int \tan x dx = -\ln|\cos x| + c, \quad \int \ln x dx = x \ln x - x + c, \quad x > 0$$

$$\int \frac{dx}{\sin x} = \ln \left| \tan \left(\frac{x}{2} \right) \right| + c \quad x \neq (2k+1)\pi \text{ and } k \in \mathbb{Z}$$

$$\int \frac{dx}{\cos x} = \ln \left| \tan \left(\frac{x}{2} + \frac{\pi}{4} \right) \right| + c \quad x \neq \frac{\pi}{2} + 2k\pi \text{ and } k \in \mathbb{Z}$$

$$\int \frac{dx}{1-x^2} = \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| + c \quad |x| \neq 1 \quad (\text{ or } \operatorname{argth} x \text{ if } x \in]-1, 1[$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \operatorname{Arcsin} x = -\operatorname{Arccos} x + c \quad |x| < 1$$

$$\int \frac{dx}{1+x^2} = \arctan x + c$$

$$\int \frac{dx}{\sqrt{x^2+1}} = \ln|x + \sqrt{x^2+1}| + c \quad (\text{ or } \operatorname{argsh} x)$$

$$\int \frac{dx}{\sqrt{x^2-1}} = \ln|x + \sqrt{x^2-1}| + c \quad |x| > 1 \quad (\text{ or } \operatorname{argch} x \text{ if } x > 1)$$

$$\int \frac{dx}{\sqrt{x^2+h}} = \ln|x + \sqrt{x^2+h}| + c \quad x^2 + h > 0$$

$$\int \frac{dx}{\operatorname{sh} x} = \ln \left| \operatorname{th} \left(\frac{x}{2} \right) \right| + c \quad x > 0 \quad \text{ or } \quad x < 0$$

$$\int \frac{dx}{\operatorname{ch} x} = 2 \operatorname{Arctan}(e^x) + c$$

$$\int \frac{dx}{\operatorname{th} x} = \ln|\operatorname{sh} x| + c \quad x > 0 \quad \text{ or } \quad x < 0$$

2. Compute the following primitives using the integration by parts

$$\int x^2 \ln x \, dx \quad \int \ln x \, dx \quad \text{then} \quad \int (\ln x)^2 \, dx, \quad \int \cos x \exp x \, dx$$

3. Calculate

$$A = \int \frac{\sinh(x)}{1+4 \cosh^2(x)} dx, \quad B = \int \sqrt{\cos(x)} \sin(x) dx$$

$$C = \int \frac{1}{\sqrt{1-x^2} \arcsin(x)} dx, \quad D = \int \frac{1}{x \cdot \ln(x) \cdot \ln(\ln(x))} dx$$

4. Determine $A = \int \frac{2x+1}{x^2+x+1} dx$, $B = \int_0^2 \frac{x}{x^4+16} dx$, $C = \int (x+1) \sinh(x) dx$, $D = \int \frac{x^3}{\sqrt{x^2+1}} dx$

5. Calculate $I_1 = \int \cos(\ln(x)) dx$, $I_2 = \int \arg \cos(x) dx$, $I_3 = \int x \cdot \tan^2(x) dx$

6. Determine $I_n = \int \frac{dt}{(t^3+1)^n}$, $I = \int \frac{dx}{1+\cos(x)}$

7. Compute the following primitive using a change of variable

$$\int (\cos x)^{2024} \sin x \, dx \quad \int \frac{1}{x \ln x} \, dx \quad \int \frac{1}{3+\exp(-x)} dx \quad \int \frac{1}{\sqrt{4x-x^2}} dx$$

8. Compute the following primitives, and precise if necessary the interval of validity of your computations :

$$(a) \quad I = \int \frac{1}{x^2+2x+5} dx, \quad I = \int \frac{1}{x^2+3x+1} dx, \quad I = \int \frac{3x-2}{5x^2-3x+2} dx, \quad I = \int \frac{6x^4-5x^3+4x^2}{2x^2-x+1} dx.$$

$$(b) \quad \int \frac{x+2}{x^2-3x-4} dx \quad \int \frac{x-1}{x^2+x+1} dx, \quad \int \sin^8 x \cos^3 x dx, \quad \int \frac{1}{\sin x} dx, \quad \int \frac{3-\sin x}{2 \cos x+3 \tan x} dx$$

9. Calculate

$$(a) \quad I = \int (x + \sqrt{x}) dx, \quad I = \int \frac{dx}{\sqrt[4]{x}}, \quad I = \int \left(\frac{3}{\sqrt{x}} - \frac{x\sqrt{x}}{4} \right) dx.$$

10. Compute the following integrals :

$$(a) \quad I = \int \frac{dx}{(\sin 3x)^2}, \quad I = \int \frac{dx}{5-2x}, \quad I = \int (\sin x)^2 \cos x dx,$$

$$(b) \quad I = \int \frac{\sin 2x}{(1+\cos 2x)^2} dx, \quad I = \int \frac{\arctan x}{1+x^2} dx, \quad I = \int \frac{1}{\sqrt{1-x^2} \arcsin x} dx,$$

$$(c) \quad I = \int \frac{\cos(\ln x)}{x} dx, \quad I = \int \frac{e^x}{\sqrt{1-e^{2x}}} dx, \quad I = \int \frac{1}{1+2x^2} dx,$$

$$(d) \quad I = \int \frac{1}{4-9x^2} dx, \quad I = \int \frac{1}{\sqrt{1-3x^2}} dx, \quad I = \int \frac{1}{2(\sin x)^2+3(\cos x)^2} dx.$$

11. Compute the following integrals :

$$(a) \quad I = \int \arcsin x dx, \quad I = \int x \arctan x dx, \quad I = \int \frac{x \arcsin x}{\sqrt{1-x^2}} dx.$$

12. Calculate primitives of type $\int \frac{Ax+B}{\sqrt{ax^2+bx+c}} dx$

$$I = \int \frac{1}{\sqrt{2-3x-4x^2}} dx, \quad I = \int \frac{1}{\sqrt{1+x+x^2}} dx, \quad I = \int \frac{x+3}{\sqrt{4x^2+4x+3}} dx, \quad I = \int \frac{x+3}{\sqrt{3+4x-4x^2}} dx.$$

13. Calculate integral of

$$(a) \quad I = \int \frac{1}{(x-1)^2(x-2)} dx, \quad I = \int \frac{x^4}{x^3+2x^2-x-2} dx, \quad I = \int \frac{2x^2-3x-3}{x^3-3x^2+7x-5} dx,$$

$$(b) \quad I = \int \frac{x^4+1}{(x^3+5x^2+7x+3)(x^2+1)} dx, \quad I = \int \frac{1}{(x^2-x)(x^2-x+1)^2} dx.$$

14. Compute

$$(a) \quad I = \int \frac{\sqrt{x^3-3x}}{6\sqrt[4]{x}} dx, \quad I = \int \sqrt{\frac{1-x}{1+x}} \frac{1}{x} dx,$$

$$(b) \quad I = \int \sqrt{\frac{1-x}{1+x}} \frac{1}{x^2} dx, \quad I = \int \sqrt{\frac{2+3x}{x-3}} dx.$$

15. Calculate primitives of type $\int R(x, \sqrt{ax^2+bx+c})$

$$I = \int \frac{1}{x-\sqrt{x^2-1}} dx, \quad I = \int \frac{1}{(2x-x^2)\sqrt{2x-x^2}} dx, \quad I = \int \frac{1}{(1+x)\sqrt{1+x+x^2}} dx.$$

16. Determine integral of

$$(a) \quad I = \int \frac{1}{5-3 \cos x} dx, \quad I = \int \frac{\sin x}{1+\sin x} dx,$$

$$(b) \quad I = \int (\cos x)^2 dx, \quad I = \int (\operatorname{tg} x)^4 dx.$$